Section - A

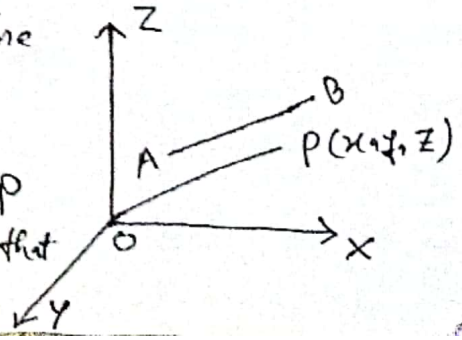
Co-ordinate Geometry and Vectoranalysis

1.(a) Define direction cosines and direction ratios. If l, m, n are direction cosines of a line, show that $l^2 + m^2 + n^2 = 1$

Solution: Direction cosines: If a given line AB makes angles α, β, γ with axes of X, Y and Z and respectively then $\cos\alpha$, $\cos\beta$ and $\cos\gamma$ are called the direction cosines of the line AB and are generally denoted by the letters l, m and n respectively i.e; $l = \cos\alpha$, $m = \cos\beta$ and $n = \cos\gamma$

Direction ratios: Consider a line AB. Then any three numbers a, b and c which are proportional to the direction cosines l, m, n respectively of the given line are called direction ratios of the given line.

2nd part Solution:- Let AB be the line whose direction cosines are l, m, n. Through the origin O draw a line OP parallel to AB and of length r so that the direction cosines of the line



OP are also l, m, n. Now if (x, y, z) be the co-ordinates of the point P then we know that

$$x = lr \;,\; y = mr \;,\; z = nr$$

Squaring and adding,

$$x^2+y^2+z^2 = l^2r^2 + m^2r^2 + n^2r^2$$

$$= r^2(l^2+m^2+n^2) \quad \text{--- (1)}$$

But $x^2+y^2+z^2 = OP^2 = r^2$ [length]

Hence from (1) we get,

$$r^2 = r^2(l^2+m^2+n^2)$$

$$\Rightarrow l^2+m^2+n^2 = 1$$

$$\Rightarrow \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

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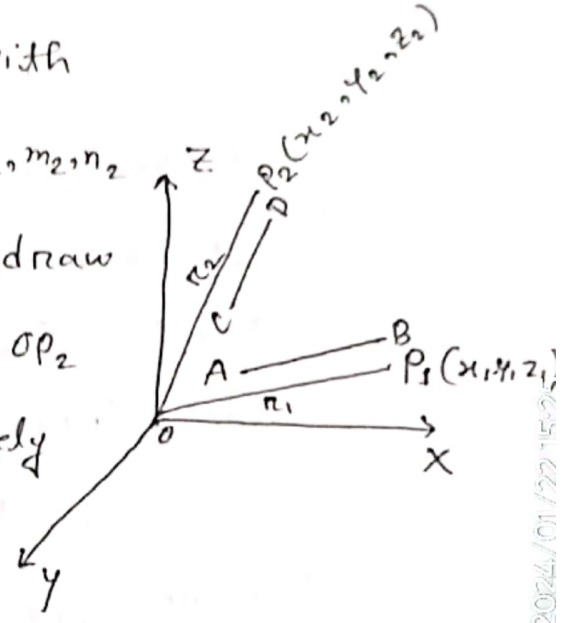
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(9)

(b) Show that $\cos\theta = l_1l_2 + m_1m_2 + n_1n_2$. If l_1, m_1, n_1 and l_2, m_2, n_2 be the direction cosines of any two lines AB and CD and θ be the angle between them. then $\cos\theta = l_1l_2 + m_1m_2 + n_1n_2$

Solution:- Given lines AB and CD with direction cosines l_1, m_1, n_1 and l_2, m_2, n_2 respectively. Through the origin O draw two lines OP_1 of length r_1 and OP_2 of length r_2 parallel respectively to the lines AB and CD. Then



$x_1 = l_1r_1$, $y_1 = m_1r_1$, $z_1 = n_1r_1$

and $x_2 = l_2r_2$, $y_2 = m_2r_2$ $z_2 = n_2r_2$

If θ be the angle between the given lines, then θ is also equal to the angle between OP_1 and OP_2

Now the projection of line OP_2 joining $(0,0,0)$ and (x_2, y_2, z_2) on the line OP_1 whose direction cosines are (l_1, m_1, n_1) is

$$(x_2-0)l_1 + (y_2-0)m_1 + (z_2-0)n_1$$

$$= l_1x_2 + m_1y_2 + n_1z_2$$

But the projection of OP_2 on OP_1 is $OP_2\cos\theta$

Therefore,

$$OP_2 \cos\theta = l_1 x_2 + m_1 y_2 + n_1 z_2$$

$$\text{or } r_2 \cos\theta = l_1 (l_2 r_2) + m_1 (m_2 r_2) + n_1 (n_2 r_2)$$

$$\therefore \cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

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(c) A variable plane is at a constant distance $3p$ from the origin and meets the axes in A, B and C. Show that the locus of the centroid of the triangle ABC is

$$x^{-2} + y^{-2} + z^{-2} = p^{-2}$$

Solution: Let the equation of the plane be

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

$$\Rightarrow \frac{x}{a} + \frac{y}{b} + \frac{z}{c} - 1 = 0 \quad \text{---} \quad (1)$$

If its distance from the origin is $3p$, then

$$3p = \frac{0+0+0-1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}$$

$$\Rightarrow \frac{1}{9p^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \quad \text{---} \quad (2)$$

If (x, y, z) be the centroid of the triangle ABC where the points are $A(a,0,0)$ $B(0,b,0)$ $C(0,0,c)$

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(3)

then $x = \frac{a+0+0}{3}$

$\Rightarrow x = \frac{a}{3}$

or $a = 3x$

Similarly, we get

$y = \frac{b}{3}$ or $b = 3y$

$z = \frac{c}{3}$ or $c = 3z$

Now putting the values of a, b, c in (2) we get the locus of the centroid of the triangle which is

$$\frac{1}{9p^2} = \frac{1}{9x^2} + \frac{1}{9y^2} + \frac{1}{9z^2}$$

$$\Rightarrow x^{-2} + y^{-2} + z^{-2} = p^{-2}$$

(Showed)

2. (a) Find the shortest distance and equation of shortest distance : $\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$; $\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}$

Find also its equation and the points where it intersects the lines

Solution :- Given lines,

$$\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1} \quad ---(1)$$

$$\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1} \quad ---(2)$$

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Let the shortest distance meet the given lines in L and M.

Any point on (1) suppose L is $(r+3, -2r+5, r+7)$

Any point on (2) suppose M is $(7r'-1, -6r'-1, r'-1)$

Direction ratios of LM are $[(x_2-x_1, y_2-y_1, z_2-z_1)]$

$(7r'-r-4, -6r'+2r-6, r'-r-8)$ ---(3)

Since LM is perpendicular to both the given lines, then

$$1(7r'-r-4)-2(-6r'+2r-6)+1(r'-r-8)=0$$

$$7(7r'-r-4)-6(-6r'+2r-6)+1(r'-r-8)=0$$

Solving this we get, $r=0$, $r'=0$

Therefore, the points where the shortest distance intersects the lines (1) and (2) i.e L is $(3,5,7)$ and M is $(-1,-1,-1)$

Now the length of the shortest distance is

$$LM = \sqrt{(-1-3)^2+(-1-5)^2+(-1-7)^2}$$

$$= \sqrt{16+36+64}$$

$$= \sqrt{116} = 2\sqrt{29}$$

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Again, direction ratios of LM are 4, 6, 8 or 2, 3, 4

[From (3), putting $n = n' = 0$]

Hence the equation of the shortest distance i.e LM is $\frac{x-3}{2} = \frac{y-5}{3} = \frac{z-7}{4}$

L(3, 5, 7)
DRs of LM
(2, 3, 4)
∴ Equation of SD
$\frac{x-3}{2} = \frac{y-5}{3} = \frac{z-7}{4}$

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(b) A plane passes through a fixed point (a, b, c) and cut the axes in A, B, C. Show that the locus of the centre of the sphere OABC is

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1$$

Solution: Let the plane be

$$\frac{x}{m} + \frac{y}{n} + \frac{z}{p} = 1 \quad \text{--- (1)}$$

if it passes through (a, b, c) then

$$\frac{a}{m} + \frac{b}{n} + \frac{c}{p} = 1 \quad \text{--- (2)}$$

Therefore it meets the axes in

A (m, 0, 0), B (0, n, 0) and C (0, 0, p)

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Also planes through A, B and C are parallel respectively to the co-ordinate planes are

$$x = m, \quad y = n, \quad z = p \quad -- (3)$$

respectively.

Thus the locus of the point of intersection of the planes is obtained by eliminating the variables m, n, p

From (2) and (3), we get.

$$\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1$$

$$\Rightarrow ax^{-1} + by^{-1} + cz^{-1} = 1 \quad \text{(Showed)}$$

(a) Find the angle between the lines:

$x - 2y + z = 0$, $x + y - z = 3$ and

$x + 2y + z = 5$, $8x + 12y + 5z = 0$

Solution:

3. (a) Define scalar product and Dot product of vectors. Determine a unit vector perpendicular to the plane of $\vec{A} = 2\hat{i} - 6\hat{j} - 3\hat{k}$ and $\vec{B} = 4\hat{i} + 3\hat{j} - \hat{k}$

Solution:- Dot or Scalar product:- The dot or scalar product of two vectors $\bar{A}$ and $\bar{B}$ denoted by $\bar{A}.\bar{B}$, is defined as the product of the magnitude of $\bar{A}$ and $\bar{B}$ and the cosine of the angle between them such that

$$\bar{A}.\bar{B} = AB\cos\theta \qquad 0 \leq \theta \leq \pi$$

2nd part Solution:- Given vectors, $\bar{A} = 2\hat{i} - 6\hat{j} - 3\hat{k}$ $B = 4\hat{i} + 3\hat{j}$

$A \times B$ is a vector perpendicular to the plane of A and B

$$A \times B = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -6 & -3 \\ 4 & 3 & 1 \end{vmatrix}$$

$$= \hat{i}(-6+9) - \hat{j}(2+12) + \hat{k}(6+24)$$

$$= 15\hat{i} - 10\hat{j} + 30\hat{k}$$

A unit vector parallel to $A \times B$ is $\dfrac{A \times B}{|A \times B|}$

(6)

$$= \frac{15\hat{i} - 10\hat{j} + 30\hat{k}}{\sqrt{(15)^2 + (-10)^2 + (30)^2}}$$

$$= \frac{3}{7}\hat{i} - \frac{2}{7}\hat{j} + \frac{6}{7}\hat{k} \quad (\text{Ans})$$

Another unit vector opposite in direction is $\frac{(-3\hat{i} + 2\hat{j} - 6}{7}$

(b) A particle moves along the curve $x = 2t^2$, $y = t^2 - 4t$, $z = -t - 5$. Find the components of its velocity and acceleration at time $t = 1$ in the direction $\hat{i} - 2\hat{j} + 2\hat{k}$

Solution:- Given curve $x = 2t^2$, $y = t^2 - 4t$, $z = -t - 5$

$$\text{Velocity} = \frac{dr}{dt}$$

$$= \frac{d}{dt}\left[(2t^2)\hat{i} + (t^2 - 4t)\hat{j} + (-t - 5)\hat{k}\right]$$

$$= (4t)\hat{i} + (2t - 4)\hat{j} - \hat{k} \quad \text{at } t = 1$$

$$= 4\hat{i} + 2\hat{j} - \hat{k} \quad \text{at } t = 1$$

Unit vector in direction $\hat{i} - 2\hat{j} + 2\hat{k}$ is $\frac{\hat{i} - 2\hat{j} + 2\hat{k}}{\sqrt{(1)^2 + (-2)^2 + (2)^2}}$

$$= \frac{1}{3}\hat{i} - \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k}$$

Then the component in the given direction is

$$(4\hat{i} - 2\hat{j} - \hat{k})\left(\frac{1}{3}\hat{i} - \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k}\right) = 2$$

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$$\text{Acceleration} = \frac{d^2 r}{dt^2} = \frac{d}{dt}\left(\frac{dr}{dt}\right)$$

$$= \frac{d}{dt}\left[(4t)\hat{i} + (2t-4)\hat{j} - k\right]$$

$$= 4\hat{i} + 2\hat{j}$$

Then the component of the acceleration in the given direction is $(4\hat{i} + 2\hat{j}) \cdot \left(\frac{1}{3}\hat{i} - \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k}\right)$

$$= 0 \quad \text{(Ans:)}$$

b ৫৭ another formate :

(i) Find a unit normal to the surface $-x^2yz + 2xy^2z = 1$ at $P(1,1,1)$

Solution: Given surface, $-x^2yz + 2xy^2z = 1$

Let $Q = -x^2yz + 2xy^2z - 1$

$= 2xy^2z - x^2yz - 1$

Now, Normal vector $\nabla Q = \left(\frac{d}{dx}\hat{i} + \frac{d}{dy}\hat{j} + \frac{d}{dz}\hat{k}\right).(2xy^2z - x^2yz - 1)$

$= (2y^2z - 2xyz)\hat{i} + (4xyz - x^2z)\hat{j} + (2xy^2 - x^2y)\hat{k}$

at $P(1,1,1)$ the normal vector is

$= (2-2)\hat{i} + (4-1)\hat{j} + (2-1)\hat{k}$

$= 3\hat{j} + \hat{k}$

Now, the unit of this normal vector is

$= \frac{3\hat{j} + \hat{k}}{\sqrt{9+1}}$

$= \frac{3}{\sqrt{10}}\hat{j} + \frac{1}{\sqrt{10}}\hat{k}$ (Ans:)

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4. (a) prove that, $\nabla^2\left(\frac{1}{r}\right) = 0$

Solution:

$$\nabla^2\left(\frac{1}{r}\right) = \left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\left(\frac{1}{\sqrt{x^2+y^2+z^2}}\right)$$

$$\frac{\partial}{\partial x}\left(\frac{1}{\sqrt{x^2+y^2+z^2}}\right) = \frac{\partial}{\partial x}(x^2+y^2+z^2)^{-1/2}$$

$$= -x(x^2+y^2+z^2)^{-3/2}$$

$$\frac{\partial^2}{\partial x^2}\left(\frac{1}{\sqrt{x^2+y^2+z^2}}\right) = \frac{\partial}{\partial x}\left[-x(x^2+y^2+z^2)^{-3/2}\right]$$

$$= 3x^2(x^2+y^2+z^2)^{-5/2} - (x^2+y^2+z^2)^{-3/2}$$

$$= \frac{2x^2-y^2-z^2}{(x^2+y^2+z^2)^{5/2}}$$

Similarly,

$$\frac{\partial^2}{\partial y^2}\left(\frac{1}{\sqrt{x^2+y^2+z^2}}\right) = \frac{2y^2-z^2-x^2}{(x^2+y^2+z^2)^{5/2}}$$

and $$\frac{\partial^2}{\partial z^2}\left(\frac{1}{\sqrt{x^2+y^2+z^2}}\right) = \frac{2z^2-x^2-y^2}{(x^2+y^2+z^2)^{5/2}}$$

Then, by addition,

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\left(\frac{1}{\sqrt{x^2+y^2+z^2}}\right)$$

$$= 0$$

The equation $\nabla^2\phi = 0$ is called Laplaces equation. It follows that $\phi = 1/r$ is.

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(b) Evaluate $\iint_S A \cdot n \, ds$, where $\vec{A} = 18z\hat{i} - 12\hat{j} + 3y\hat{k}$ and S is that part of the plane $2x + 3y + 6z = 12$ which is located in the first octant.

Solution: The surface S and its projection R on the xy-plane are shown.

$$\iint_S A \cdot n \, ds$$

$$= \iint_R A \cdot n \frac{dx\,dy}{|n \cdot k|}$$

To obtain n, note that a vector perpendicular to the surface $2x + 3y + 6z = 12$ is given by $\nabla(2x + 3y + 6z) = 2\hat{i} + 3\hat{j} + 6\hat{k}$. Then a unit normal to any point of S

$$n = \frac{2\hat{i} + 3\hat{j} + 6\hat{k}}{\sqrt{2^2 + 3^2 + 6^2}}$$

$$= \frac{2}{7}\hat{i} + \frac{3}{7}\hat{j} + \frac{6}{7}\hat{k}$$

Thus $n \cdot k = \left(\frac{2}{7}\hat{i} + \frac{3}{7}\hat{j} + \frac{6}{7}\hat{k}\right) \cdot k$

$$= \frac{6}{7} \text{ and so}$$

$$\frac{dx\,dy}{|n \cdot k|} = \frac{7}{6} dx\,dy$$

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Also $A \cdot n = (18z\,i - 12\,j + 3y\,k) \cdot \left(\frac{2}{7}i + \frac{3}{7}j + \frac{6}{7}k\right)$

$$= \frac{36z - 36 + 18y}{7}$$

$$= \frac{36 - 12x}{7}$$

using the fact that $z = (12 - 2x - 3y)/6$ from the equation of S. Then

$$\iint_S A \cdot n\, ds = \iint_R A \cdot n \frac{dx\,dy}{|n \cdot k|}$$

$$= \iint_R \left(\frac{36 - 12x}{7}\right) \frac{7}{6}\, dx\,dy$$

$$= \iint_R (6 - 2x)\, dx\,dy$$

To evaluate this double integral over R. keep x fixed x fixed and integrate with respect to y from $y = 0$ to $y = (12 - 2x)/3$. Then integrate with respect to x from $x = 0$ to $x = 6$. In this manner R is completely covered. The integral becomes

$$\int_{x=0}^{6} \int_{y=0}^{(12-2x)/3} (6 - 2x)\, dy\, dx = \int_{x=0}^{6} \left(24 - 12x + \frac{4x^2}{3}\right) dx$$

$$= 24$$

(c) Verify Greens theorem in the plane for $\oint_C (xy + y^2)\,dx + x^2\,dy$ where C is the closed curve of the region bounded by $y = x$ and $y = x^2$

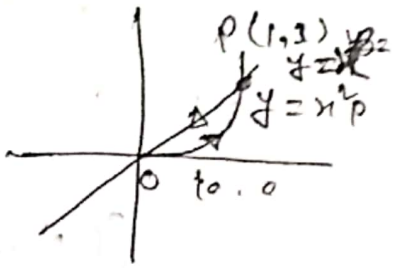
Solution: Given $y = x$ and $y = x^2$

$\therefore Vx = x^2$

$x(x^2 - x) = 0$

$x(x-1) = 0$

$\therefore x = 0$ or $x = 1$



when $x = 0$ then $y = 0$

when $x = 1$ then $y = 1$

Therefore the given curve interset of $(0,0)$ and $(1,1)$ the positive direction in traversing C i. as shown in the adjacent figure Along $y = x^2$ the line integral become,

$$\int_0^1 \{(x \cdot x^2 + x^4)\,dx + x^2 \cdot 2x\,dx\}$$

$$= \int_0^1 (x^3 + x^4 + 2x^3)\,dx$$

$$= \int_0^1 (3x^3 + x^4)\,dx$$

$$= \left[3\frac{x^4}{4} + \frac{x^5}{5}\right]_0^1$$

$$= \frac{3}{4} + \frac{1}{5}$$

$$= \frac{15 + 4}{20}$$

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$= \frac{19}{20}$

Along $y = x$ the line integral become,

$$\int_1^0 \{(x.x + x^2)dx + x^2 dx\}$$

$$= \int_1^0 (x^2 + x^2 + x^2)\, dx$$

$$= \int_1^0 3x^2 dx = \left[3.\frac{x^3}{3}\right]_1^0$$

$$= x^3$$

$$= (0 - 1)$$

$$= -1$$

Then the required line integral,

$$\frac{19}{20} - 1$$

$$= \frac{-1}{20} \quad (\text{Ans;})$$

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